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Supporting multi-depot and stochastic waste collection management in clustered urban areas via simulation–optimization

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Waste collection is one of the most critical logistics activities in modern cities with considerable impact on the quality of life, urban environment, city attractiveness, traffic flows and municipal budgets. Despite the problem's relevance, most existing work addresses simplified versions where container loads are considered to be known in advance and served by a single vehicle depot. Waste levels, however, cannot be estimated with complete certainty as they are only revealed at collection. Furthermore, in large cities and clustered urban areas, multiple depots from which collection routes originate are common, although cooperation among vehicles from different depots is rarely considered. This paper analyses a rich version of the waste collection problem with multiple depots and stochastic demands by proposing a hybrid algorithm combining metaheuristics with simulation. Our 'simheuristic' approach allows for studying the effects of cooperation among different depots, thus quantifying the potential savings this cooperation could provide to city governments and waste collection companies.
Journal of Simulation (2016). doi:10.1057/s41273-016-0002-4

Keywords: simheuristics; simulation–optimization; waste collection management; multi-depot vehicle routing problem; horizontal cooperation; stochastic demands

1. Introduction

Uncollected garbage, pollution and traffic jams through extensive vehicle usage as well as high operative costs are some of the main problems arising through sub-optimal waste collection. For this reason, waste management is among the most important municipal services, especially in urban areas facing increasing waste production and growing population numbers (Crossette *et al*, 2011; Malakahmad *et al*, 2014; Son, 2014). In an attempt to optimize urban waste collection, the waste collection problem (WCP) as essential part of managing the garbage supply chain has been addressed by different researchers in the past as special instance of the well-known vehicle routing problem (VRP) in which a vehicle fleet located at a central depot has to serve a number of customers (Beliën *et al*, 2014).

The VRP is one of the most-studied combinatorial optimization problems. Given a fleet of capacitated vehicles located at a central depot n_0, a set of n customers $I = \{n_1, n_2,\ldots,n_n\}$ with a given demand $d_i \geq 0$ at each customer $i \in I$ has to be served. The travel costs between any two nodes in $V = I \cup \{n_0\}$ (e.g. distance, time, etc.) are known and represented by c_{ij} $(i, j \in V)$. While the objective function is to minimize costs, problem constraints include that every customer is served by only one vehicle, all routes start and end at the central depot and that no vehicle can stop twice at the same customer (Toth and Vigo, 2014). Instead of delivering demand (i.e. waste levels), the aim of the WCP is to empty its clients (i.e. waste containers) according to a specific routing plan. As outlined in Figure 1, collection vehicles need to empty themselves before returning to the depot (Route 1) or once the vehicle capacity is reached (Route 2). Multiple garbage disposal trips on a single route are possible. The collected waste is disposed at one of m landfills, included in the original node-set V by adding the sub-set $L = \{n_{n+1}, n_{n+2},\ldots,n_{n+m}\}$ to the original number of nodes.

Despite the importance of managing garbage and different approaches to optimize waste collection route planning, there is a lack of scientific publications considering rich and realistic WCP environments, e.g. scenarios with a large number of clients, multiple vehicle depots and uncertainty in waste levels or demands. In order to tackle the WCP with stochastic demands (WCPSDs), this paper proposes a hybrid algorithm based on the simheuristic framework described in Grasas *et al* (2014), which has proven to be an effective simulation–optimization approach for different kinds of stochastic combinatorial optimization problems. In this paper, Monte Carlo simulation is integrated into a metaheuristic framework in order to estimate the expected cost and cost variability of

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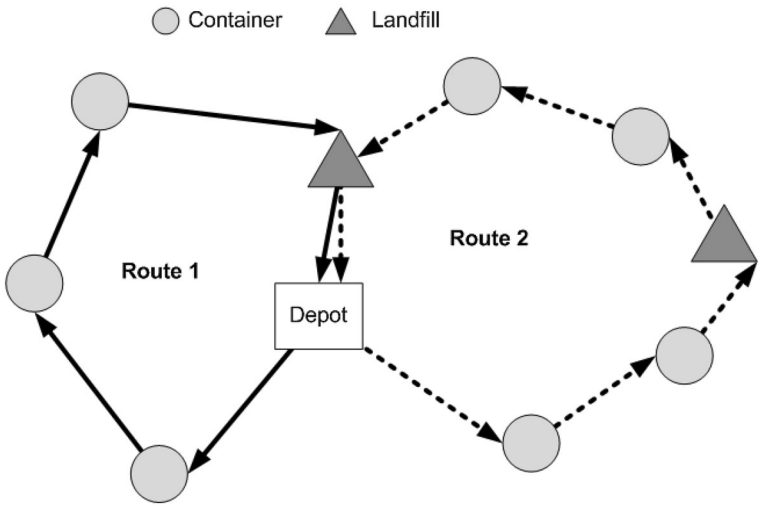


Figure 1 Example of a WCP solution with two routes.

promising routing plans initially generated for the deterministic version of the problem. Our simheuristic algorithm also makes use of biased (oriented) randomization techniques, similar to those proposed in Faulin *et al* (2008), in order to guide the random process that generates new solutions. As discussed in Juan *et al* (2013b), biased-randomized algorithms are easily parallelizable and able to provide real-time solutions to complex VRPs with stochastic demands.

As far as we know, the algorithm introduced here is the first one addressing the multi-depot WCPSDs (MDWCPSDs), which is closely related to real-life waste collection activities. Additionally, our approach allows for quantifying the potential cost savings derived from using horizontal cooperation (HC) strategies, i.e. the sharing of information and resources among different waste management service providers.

This paper is structured as follows: Section 2 provides an overview on related works regarding simulation–optimization, waste collection and HC. The algorithm to solve the MDWCPSD is introduced in Section 3, while Section 4 describes the computational experiments used to validate our findings. The obtained results are discussed in Section 5. Finally, Section 6 summarizes the highlights of this work.

2. Related studies

2.1. Combining simulation with metaheuristics

The need for advanced optimization methods to solve real-life problem settings, characterized by stochastic inputs and probability constraints, in combination with enhanced computational possibilities, has led to the development of different hybrid simulation–optimization approaches. In this context, the combination of simulation techniques with optimization methods in which simulation acts as an evaluation function for the optimization outputs has proven to be a fast, reliable and easy-to-use approach in the solution of large-sized optimization problems under uncertainty (Andradottir, 2006).

A general methodology combining metaheuristics and simulation, the so-called simheuristics, is discussed in detail by Juan *et al* (2015a, b). A particular adaptation in the case of the iterated local search (ILS) metaheuristic is given in Grasas *et al* (2014). Through a close integration between simulation and optimization, this approach has been successfully applied to different problem settings. Related to transportation and logistics, Juan *et al* (2011) test the effects on safety stocks for VRPs under uncertainty. A simheuristic approach to solve the complex inventory routing problem combining routing and inventory management decisions is suggested and analysed by Juan *et al* (2014b). Also, regarding other application fields, the combination of metaheuristics with simulation has been implemented, for example, in production planning (Figueira *et al*, 2013), stochastic scheduling (Juan *et al*, 2014a), or in the improvement of Internet computing services (Cabrera *et al*, 2014).

2.2. The waste collection problem

The WCP formulated as an optimization problem has been widely discussed in the scientific literature. In an extensive literature review on nearly 80 publications on garbage collection since 1970, Beliën *et al* (2014) summarize theoretical and practical research work on the WCP as location and routing problems, classified according to different dimensions (e.g. type of waste or solution methodologies). Concerning the WCP as a routing problem in the collection of household waste, an insertion metaheuristic for large-sized (deterministic) WCP settings with a single depot and multiple landfills is presented by Kim *et al* (2006), while Benjamin and Beasley

(2010) combine tabu with variable neighbourhood search. Further concerning the solution of deterministic WCPs, Buhrkal *et al* (2012) suggest an adaptive large neighbourhood search metaheuristic with destroy-and-repair techniques.

The WCPSDs with small customer sets based on a case study in Malaysia is addressed by Ismail and Loh (2009) by applying an ant colony optimization metaheuristic. After defining an initial a priori solution, waste levels are modelled using a discrete probability distribution. Necessary recourse action to the initially defined collection route is then included when necessary. A similar approach to the same problem (with up to 50 customers) is solved by Ismail and Irhamah (2008) using a hybrid approach based on a genetic algorithm and tabu search.

2.3. Multi-depot VRPs

VRPs with multiple depots (MDVRPs) have been subject to research for many decades. Hereby, customers are allocated to different depots in a combinatorial assignment problem, before each depot–node assignment combination is solved as VRP. Pisinger and Ropke (2007) propose an adaptive large neighbourhood search metaheuristic which they test on variance instance obtaining promising results, while requiring the fine-tuning of as much as 14 parameters. Vidal *et al* (2012) use a hybrid genetic algorithm to solve different VRP variants, among them the MDVRP. The integration of biased (oriented) randomization techniques inside an ILS framework designed to solve routing problems with multiple depots is discussed in Juan *et al* (2015b). For more general overviews over solution methods to the MDVRP, the reader is addressed to Kumar (2012) and Karakatič and Podgorelec (2015).

While the MDVRP is a widely studied problem, the literature is scarce for the special case of waste collection routing, even non-existent for the MDWCPSD. A closely related problem setting to the MDWCP is discussed by Crevier *et al* (2007), who apply an adaptive memory principle combined with integer programming and tabu search metaheuristics to consider the MDVRP with inner-route replenishment points. Hemmelmayr *et al* (2011) consider multiple depots and landfills in a case study for node routing for the collection of waste from public waste delivery points to landfills by applying a hybrid approach. Combining variable neighbourhood search and dynamic programming, the authors solve smaller problem instances with up to 75 collection points. Additionally, Ramos *et al* (2014) improve the economic and environmental impacts of recyclable waste collection by solving the WCP with multiple depots as a mixed integer linear programming model.

2.4. Horizontal cooperation

In many supply chains from different sectors, HC is being discovered as a tool to reduce overall transportation costs, vehicle usage and the environmental impact of transportation activities. HC is defined by Bahinipati *et al* (2009) as "business agreement between two or more companies at the same level in the supply chain or network in order to allow ease of work and cooperation towards achieving a common objective". By sharing information and facilities, companies hope for the effects of economies of scale through better resource usage. In this context, HC yields several potential applications concerning the development of sustainable logistics networks in different sectors, e.g. by public service providers (Ballot and Fontane, 2010; Joyce and Drumaux, 2014; Perez-Bernabeu *et al*, 2015).

A simple example of the benefits of HC in waste collection can be seen in Figure 2. In the non-cooperative scenario shown on the left, waste management service providers (each represented by a single depot) do not share information about waste containers to be emptied and plan their respective collection routes individually. It can be intuitively observed that the cooperative scenario (on the right) in which containers, landfills and depots are shared leads to lower routing costs, less traffic, higher service levels and a lower environmental impact through the waste transportation activities. Applications of HC in waste collection can be thought of especially in large clustered cities with various waste management service providers responsible for different areas of a municipality and highly clustered metropolitan areas (e.g. the Ruhr area in Germany or greater Barcelona).

3. Overview of our simulation–optimization algorithm

Our methodology to tackle the complex MDWCPSD is based on the SimILS general framework described in Grasas *et al* (2014). This is a simheuristic framework combining biased (oriented) randomization, the ILS metaheuristic and Monte Carlo simulation. Thus, at different stages of the algorithm, Monte Carlo simulation is hereby applied to consider the performance of a predefined deterministic solution in a stochastic environment, as will be explained below in more detail. The rest of this section provides additional details of our approach and on the particular simheuristic algorithm proposed.

3.1. Simheuristics to consider WCP uncertainty

In contrast to a deterministic WCP setting in which input variables are considered to be known during the route planning phase, actual garbage loads to collect are only revealed once a collection vehicle reaches a pick-up point in reality. Given a WCPSD, the first step of the simheuristic procedure is to transform the stochastic problem setting into its deterministic counterpart. That is, the expected waste levels of each container on a waste collection route are used as deterministic input variables. This deterministic WCP is then solved using an efficient optimization approach. Therefore, we apply a randomized version of the well-known Clarke-and-Wright savings heuristic, as discussed in Juan *et al* (2013a). By

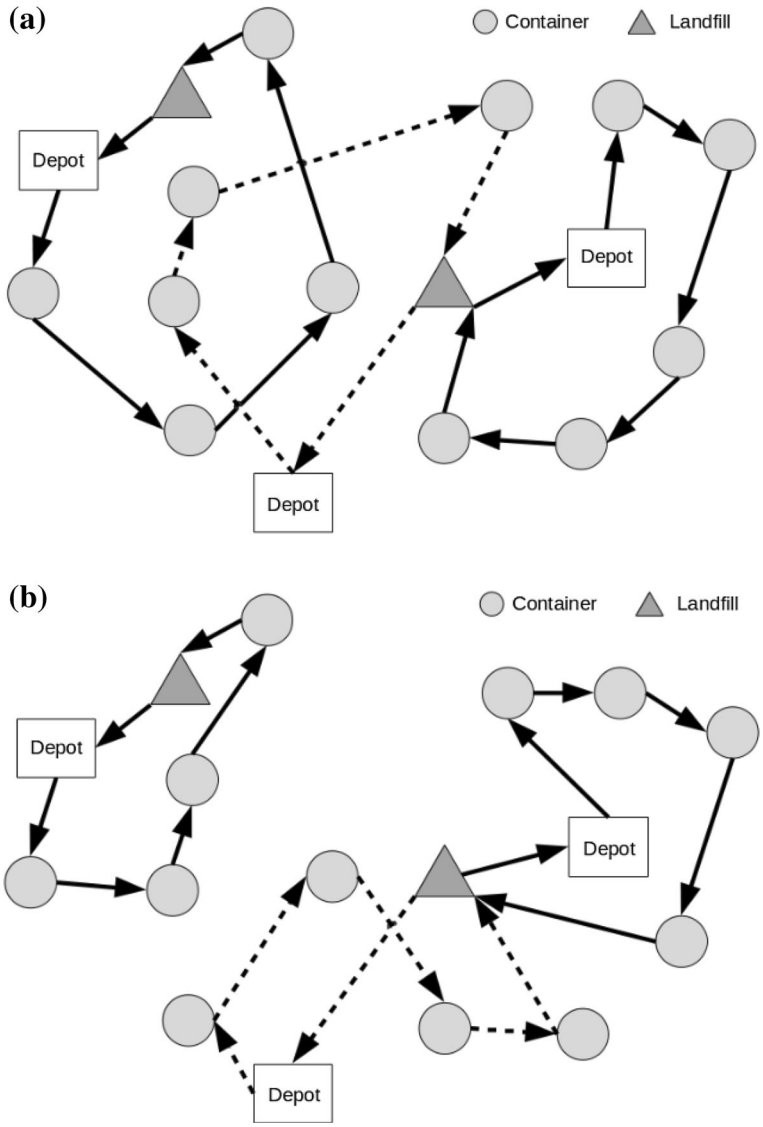


Figure 2 Non-cooperative (**a**) and cooperative (**b**) scenarios in waste management.

adapting this multi-start routing algorithm to the WCP, a number of predefined (a priori) solutions are constructed during detSolTime seconds. This procedure is shown in Algorithm 1.

After the generation of multiple solutions, a set of d deterministic solutions is tested for its behaviour in a stochastic environment by repeatedly sampling random waste levels using Monte Carlo simulation (Raychaudhuri, 2008). In order to keep computational efforts reasonable, not all deterministic solutions are tested at this point, but only the most 'promising' deterministic solutions are sent to the simulation stage. Hereby, the waste levels of each container to be served are modelled using a log-normal distribution with waste level variance Var and the expected waste level at each container as mean. During nIterSim (usually several hundred/thousands) simulation runs, a tested predefined solution is completed considering the uncertainty of pick-up amounts by only revealing the actual waste level at a container once a collection vehicle reaches it. Hereby, the order of containers to visit is not changed. However, limited vehicle capacities will lead to route failures when collected garbage exceeds the initially planned amount, making additional landfill trips necessary. Our solution algorithm penalizes route failures with an additional round trip to the closest landfill, before the planned route is continued.

```
01 procedure generateEliteSolutions (Map)
02   S ← {}              // set of elite solutions
03   while !terminate(nDetSol, detSolTime) do
04     sol ← solveBiasedRandomizedCWS()
05     if isEliteSolution(sol, d) then       //save d best deterministic sol.
06       S.add(sol)
07     end if
08   end while
09   for i ← 1 to d do    //Run short simulation on promising sol
10     evaluateReliability(S_i, Var, nIterShort)
11   end for
12   S ← resortSolutions(S)
13   return S     //return elite solutions
14 end procedure
```

Algorithm 1: Generation of solutions.

By summing up total route failure costs in all simulation runs and dividing them by nIterSim, an unbiased estimation of expected route failures can be obtained. At this stage, it often happens that the most promising deterministic WCP solutions turn out to be less competitive in a stochastic environment due to high expected additional routing costs. Besides expected route failure costs as quality indicator of the predefined route under uncertainty, the described procedure allows estimating the solution reliability by considering the proportion of routes in which a route failure occurs during the simulation runs. Finally, the solutions are re-ranked according to the total of routing and expected route failure costs. If necessary at this point, a longer simulation run can be conducted with the most promising stochastic solutions to get more reliable results of a number of stochastic elite solutions.

3.2. Combination of simulation with oriented randomization and ILS

In order to consider multiple depots in the planning of waste collection routes with stochastic waste levels, we integrate oriented randomization (Faulin *et al*, 2008) and ILS at different stages of the simulation–optimization approach, as shown in Algorithm 2. Consider a WCP with multiple depots, the first step in the solution process is to solve the node–depot assignment problem, defining which waste container is emptied from which depot. For this purpose, a priority list of containers for each depot is constructed according to a distance-based criterion. After calculating the distance of each waste container to each depot $k \in V_d$, a priority list based on the marginal savings μ_i^k of emptying container i from depot k compared to serving it from the best alternative depot k^*, such that $\mu_i^k = c_i^{k^*} - c_i^k$, is built. Next, the nodes are randomly assigned to the depots according to a round-robin criterion. Hereby, the depots iteratively 'choose' a (unassigned) container to serve.

As suggested in Juan *et al* (2015b), the process of assigning nodes to a depot is randomized but using a skewed distribution instead of a uniform one. After sorting the priority list of containers, a skewed geometric distribution is applied in such a way that containers with a higher position in the priority list are more likely to be selected by the respected depot (thus making the randomized selection process 'biased' or 'oriented'). The exact probabilities of each container to be included in a specific node–depot collection map are defined through a distribution parameter p ($0 < p < 1$). This randomized process allows the construction of nMaps different node–depot allocation maps. We evaluate the competitiveness of each map by solving the deterministic WCP setting for depot and its assigned nodes, before the WCP costs for each depot are summed up to obtain a preliminary MDWCP solution. After all maps have been constructed, the d most promising node–depot allocations are defined.

With each promising solution, a short simulation run with nIterShort iterations is started next to evaluate the performance of the predefined routing plan in a stochastic environment. Based on the deterministic routing and route failure costs, a stochastic initial solution is defined and used as an initial upper limit (currentBest) in the ILS process started in the following. Based on the work of Grasas *et al* (2014), the node–depot allocation maps are hereby perturbed by applying a destroy-and-repair method to the current best solution during time-based termination criteria (maxTimeILS). During each

perturbation, p^* percent of containers are exchanged between different node–depot allocations before a new short simulation is performed to assess the quality of the newly constructed map and compare it to the current best.

Whenever a new best solution considering total costs is found, the new solution is listed as promising (in promisingSolsStoch) and replaces the current best. In order to escape local minima in the solution search, an acceptance criterion is implemented allowing a solution worsening if the last iteration from x to x' [such that $f(x') < f(x)$] was an improvement and the difference between the current best and new solution is not greater than the latest improvement ($|f(x'') - f(x')| < f(x) - f(x')$). Once the ILS is finished, the m best solutions in promisingSolsStoch undergo a long simulation run to define the expected route failure costs and the solution reliabilities more closely. Finally, a list of elite MDWCPSD with the respective total costs and solution reliabilities are returned by the process.

simulation process. Furthermore, we consider that each landfill is a disposal site and a vehicle depot at the same time, leading to different MDWCP instances. In order to limit depot capacities, not more than maxDemand percent of total waste to be collected is assigned to a single depot, in order to keep the workload balanced and limit the vehicles per depot. Time windows for the waste containers and vehicle depots are not considered in this work.

The MDWCP can be seen as a cooperative scenario in which different waste management service providers apply HC in which containers, depots and landfills are shared to reduce garbage collection costs. We quantify the benefits of HC by comparing the MDWCPSD costs to a non-cooperative scenario in which each depot serves its own customers and no resources are shared. For each depot of a problem instance, nodes are assigned through a greedy round-robin criterion to create a number of single-depot WCPs. A priority list of nodes based on marginal savings (as

```
01  Input: Vc; I; L          //Depots, customers and landfills
02  S ← {}                   //Set of elite solutions
03  D ← {}                  //Set of depot-node allocation maps
04  D ← generateInitialMaps(nMaps)     //Generate node-depot maps
05  S ← generateEliteSolutions(D)      //Run simheuristic on each map
06  while time() < maxTImeILS do
07    D'← perturbate(D')               //perturbate elite maps
08    S'← generateEliteSolutons(D')
09    if acceptNewSolution(S,S') then
10      D ← D'          //Update current map
11      S ← S'          //Update elite solutions
12    end if
13  end while
14  for i ← 1 to d do      //Run extensive simulation on promising sol
15    evaluateReliability(Si, Var, nIterShort);
16  end for;
17  S ← resortSolutions(S)
18  return S               // Return elite solutions
```

Algorithm 2: Multi-depot simheuristic approach.

4. Computational experiments

In order to test our approach, we adopted the benchmarks provided by Kim *et al* (2006). The original set of instances considers different-sized (102–2100 nodes) deterministic WCPs with a single depot, multiple landfills and capacitated vehicles. In order to make them suitable for the MDWCPSD, we use the deterministic load levels at each container as expected waste levels and the mean of the log-normal distribution from which waste levels are modelled during the described earlier) is established. At each assignment iteration however, the current depot chooses the node with the highest savings potential to include on the planned collection route, leading to a single-depot WCP in which waste containers are assigned to vehicle depots according to a distance-based criterion. In essence, this leads to a possible node–depot allocation map with multiple depots, whereby the routing costs are estimated through the procedure described in Algorithm 1. However, as the node–depot

assignment problem is redundant in the non-cooperative scenario without HC, it is not possible to consider different container allocations and their respective routing costs.

The algorithm is implemented as Java application on a personal computer with an Intel® BYT-M 4Core™ processor with 2.66 GHz and 4 GB RAM, using the following parameters:

- *nMaps*: 100 (initial node–depot assignment maps created),
- *Var 0.05*: (waste level variance),
- *d*: 5 (promising deterministic solutions),
- *m*: 5 (promising stochastic solutions),
- *nIterSimShort*: 500 runs,
- *nIterSimLong*: 5000 runs,
- *detSolTime*: 5 s,
- *maxTimeILS*: 300 s,
- *p**: 0.3 % (nodes exchanged during perturbation),
- *p*: 0.4 (distribution parameter for biased randomization) and
- *maxDemand*: 60 %.

In Table 1, the best found solutions for the cooperative and non-cooperative scenario of each instance (and one row with the total over all tested problem settings) are listed. Column (1) shows the number of used vehicles. The opened depots are listed in column (2), before columns (3)–(5) show the fixed costs, variable (route-failure) costs and total costs, respectively. Furthermore, the percentage difference of total costs for each scenario is outlined.

5. Discussion of results

The obtained results concerning total costs, used vehicles and opened depots for the cooperative scenario (i.e. the MDWCPSD solution) and the non-cooperative counterpart are outlined in Figure 3. Concerning the total expected costs, the routing costs of each instance can be improved, with an average gap of −12% over all instances. Next to possible route savings, HC leads to less used vehicles (91 compared to 105 in the non-cooperative scenario over all instances) and less opened depots (41–55). While the objective function of this paper was to reduce total routing costs, the number of used vehicles and opened depots can also be an important driver for HC in practice, as municipalities need to consider acquisition and maintenance costs for collection vehicles. Moreover,

Table 1 Table of results

Instances	*Scenarios*	*Vehicles (1)*	*Depots (2)*	*Fixed costs (3)*	*Variable costs (4)*	*Total costs (5)*
Kim102	Cooperative	4	3	154.46	0.00	154.46
	Non-cooperative	3	3	190.99	0.28	191.27
	%-Difference					−19
Kim277	Cooperative	2	2	495.47	17.59	513.06
	Non-cooperative	2	2	503.52	12.54	516.06
	%-Difference					−1
Kim335	Cooperative	6	3	194.99	1.38	196.37
	Non-cooperative	8	5	304.88	1.77	306.65
	%-Difference					−36
Kim444	Cooperative	11	2	74.61	0.85	75.45
	Non-cooperative	11	2	79.00	0.91	79.91
	%-Difference					−6
Kim804	Cooperative	9	9	925.02	5.57	930.59
	Non-cooperative	20	20	1579.69	3.66	1583.35
	%-Difference					−41
Kim1051	Cooperative	17	2	2133.22	55.36	2188.58
	Non-cooperative	18	3	2226.68	46.74	2273.42
	%-Difference					−4
Kim1351	Cooperative	9	4	902.00	12.80	914.80
	Non-cooperative	9	4	909.28	27.61	936.89
	%-Difference					−2
Kim1599	Cooperative	12	3	1098.30	21.57	1119.87
	Non-cooperative	13	3	1307.04	19.53	1326.57
	%-Difference					−16
Kim1932	Cooperative	9	5	1105.75	14.28	1120.03
	Non-cooperative	9	5	1119.67	17.21	1136.88
	%-Difference					−1
Kim2100	Cooperative	12	8	1560.26	9.00	1569.26
	Non-cooperative	12	8	1573.71	10.08	1583.79
	%-Difference					−1
Averages	Cooperative	91	41	8644.08	138.4	8782.47
	Non-cooperative	105	55	9794.46	140.33	9934.79
	%-Difference					−12

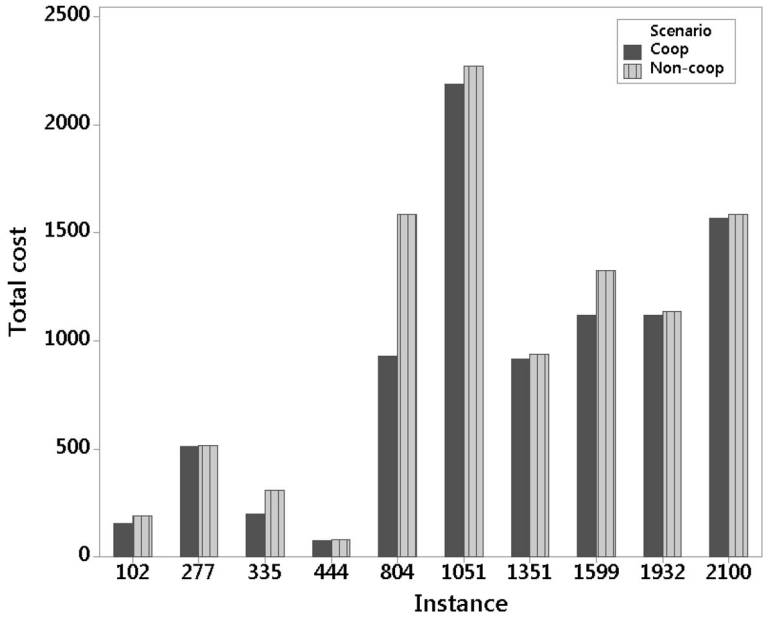


Figure 3 Comparison of total costs between cooperative and non-cooperative scenarios.

opening costs and running expenses of vehicle depots should not be neglected. Looking at the results more closely, the large differences on the positive effects of HC concerning total costs can be explained with the topology, i.e. the geographical distribution of the containers with respect to the potential depots. Obviously, instances in which only scattered depot–node allocation maps can be established tend to yield much better results in cooperative scenarios.

6. Conclusion

This paper has presented a simulation–optimization approach to support efficient waste collection management under scenarios characterized by the existence of multiple depots and stochastic demands (waste levels). Despite the fact that these scenarios are common in clustered urban areas and large cities, they had never been analysed before in the scientific literature. This is probably due to their inherent complexity, which requires the use of hybrid simulation–optimization methods. Thus, our approach relies on the combination of biased (oriented) randomization techniques, metaheuristics and Monte Carlo simulation. The proposed hybrid algorithm was tested on a set of large-sized WCP instances. Furthermore, the possible benefits of HC among different waste management service providers have been analysed. These benefits refer to savings in total costs, number of used vehicles and number of opened depots. Although it is evident that these savings are highly dependent on the specific topology and characteristics of each instance, the computational experiments show that our algorithm is able to quantify them for all the proposed benchmarks, thus supporting the idea that sharing resources (i.e. containers, landfills, depots and/or vehicles) can provide significant savings in those cases in which depots and their assigned containers are geographically scattered.

Acknowledgements—This work has been partially supported by the Spanish Ministry of Economy and Competitiveness (TRA2013-48180-C3-P and TRA2015-71883-REDT) and FEDER. Likewise, we want to acknowledge the support received from the Department of Universities, Research and Information Society of the Catalan Government (2014-CTP-00001) and the doctoral grant of the UOC.

Statement of contribution This paper presents a simulation–optimization approach to support efficient waste collection management under scenarios characterized by the existence of multiple depots and stochastic demands (waste levels). Despite the fact that these scenarios are common in clustered urban areas and large cities, they had never been analysed before in the scientific literature. This is probably due to their inherent complexity, which requires the use of hybrid simulation–optimization methods. Thus, our approach relies on the combination of biased (oriented) randomization techniques, metaheuristics and Monte Carlo simulation. The proposed hybrid algorithm was tested on a set of large-sized WCP instances. Furthermore, the possible benefits of HC among different waste management service providers are analysed. These benefits refer to savings in total costs, number of used vehicles and number of opened depots. Although it is evident that these savings are highly dependent on the specific topology and characteristics of each instance, the computational experiments show that our algorithm is able to quantify them for all the proposed benchmarks, thus supporting the idea that sharing resources (i.e. containers, landfills, depots and/or vehicles) can provide significant savings in those cases in which depots and their assigned containers are geographically scattered.

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Received 2 February 2016;
accepted 11 May 2016